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Optimization of stability-constrained geometrically nonlinear shallow trusses using an arc length sparse method with a strain energy density approach

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ABSTRACT

A technique for the optimization of stability-constrained geometrically nonlinear shallow trusses with snap-through behavior is demonstrated using the arc length method and a strain energy density approach within a discrete finite-element formulation. The optimization method uses an iterative scheme that evaluates the performance of the design variables and then updates them according to a recursive formula that is controlled by the arc length method. A minimum weight design is achieved when a uniform nonlinear strain energy density is found in all members. This minimal condition places the design load just below the critical-limit load that causes snap-through of the structure. The optimization scheme is programmed into a nonlinear finite-element algorithm to find the large strain energy at critical-limit loads. Examples of highly nonlinear trusses that are found in literature are presented to verify the method.

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1. Introduction

Large aerospace truss systems have been proposed over the years to support the future of orbital space technologies. For example, since the 1960s, various research organizations have supported the deployment of space-based solar power. The US Department of Defense is currently studying how to deploy a solar power constellation to supply power to remote bases or on the battlefield. Also during the 1960s, Rockwell International began to study large space truss structures for supporting microwave antennae and mirrors [1]. The International Space Station currently makes use of deployable trusses to support its solar array panels. All of these space-based concepts rely on truss systems to support their mission goals because a truss is the easier to assemble in microgravity and offers a minimum weight design. The lighter and more efficient that these structural systems become (i.e. with very slender members), the more likely they are to have nonlinear stability problems. The analysis challenge is to find effective techniques that go beyond the identification of Euler-type bifurcation points and find the optimal truss to carry applied limit loads without becoming unstable and releasing stored strain energy.

Weight optimization of slender truss members with stability constraints has been investigated by other researchers using various analytical techniques. Prager [2] discusses a general method of optimality that is subject to a single design constraint such as displacement, critical buckling load or natural frequency. Prager and Taylor [3] investigated columns under linear buckling constraints. Other investigators followed with schemes to optimize columns with multiple eigenvalues and design sensitivities with constraints on buckling. Some of the earliest finite-element optimization investigations began in the 1970s with linear buckling that ignored any large deformations. Khot et al. [4] presented an optimality criterion for minimizing weight to structures with stability requirements by using a linear eigenvalue buckling solution within a finite-element analysis. Several other published works have studied the nonlinear buckling of shallow trusses. Kamat and Ruangsilasingha [5] addressed the problem of maximizing the critical load of shallow, three-dimensional (3-D) trusses. They developed sensitivity derivatives of the critical load with respect to design variables. Many case studies used in previous published works were taken from Crisfield [6,7]. The case examples he used challenged many nonlinear solvers to correctly follow a nonlinear snap-through load path. Crisfield used these problems to test nonlinear behavior in truss structures and to demonstrate the importance of identifying the critical-limit load along the equilibrium path.

Imperfections in the truss geometry also can reduce the load-carrying capability of a system. An investigation of random initial imperfections for frame-type structures that exhibit instability at limit points was performed by Warren [8]. He used an arc length approach to trace the nonlinear equilibrium paths of simple benchmark examples. Khot and Kamat [9] explains that the reduction in

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load capacity results primarily from the nonlinear response of a system and secondarily from imperfections in the geometry.

Optimizing the weight of truss structures that are undergoing large displacements requires analysis methods that accurately trace the equilibrium path and identify critical-limit points. Most "shallow type" space truss designs will have snap-through response curves that are similar to those shown in Fig. 1. Depending on the geometry of the design, the structure may have a snap-back path or may respond with bifurcation buckling. These types of stability responses result from the level of nonlinearity of the geometry in the design and the applied loading. Verification problems from Crisfield [6,7,10] show the snap-through and snap-back behavior of several shallow truss designs. Achieving a minimum weight design of shallow trusses that are similar to the design examples from Crisfield requires limit-point stability constraints that can maximize critical load.

In Ref. [11], Khot et al. offer exact closed-form solutions to simple shallow truss problems that have been optimized for weight. They make use of the minimization of the total potential energy that maximizes the load-carrying capability without instability. Optimizations of geometrically nonlinear trusses that are similar to those used by Crisfield were attempted later by Khot and Kamat [9]. These examples highlight the challenges that have been encountered with many analysis techniques in following a nonlinear response path up to a critical-limit load. The sample problems also fully expose the shortcomings of modern finite-element programs by their lack of optimization scheme for stability-constrained problems. Sedaghati et al. [12] developed an optimization technique using the group theoretic approach. The method has stability constraints that use a strain energy density (SED) recurrence relationship to update the design variables. The technique involves obtaining a uniform SED in all members to establish a minimum weight design. Sedaghati et al. further made use of shallow flexible truss structures to test and verify their methods.

The method used in this paper achieves an optimal design by using the principal of SED and parameters in the arc length method to trace the nonlinear equilibrium path up to an instability point. A stable structural system exists when deformations are increasing as the applied load is increased; an unstable system exists when deformations are still increasing but the loads are decreasing.

The Hrinda optimization method used in this paper achieves an optimal design by combining the arc length method with a design-variable update scheme that identifies a uniform nonlinear SED in all members. The method uses an iterative scheme that evaluates the performance of the design variables and then updates them according to a recursive formula that is controlled by the arc length method. The method is programmed into a new finite-element algorithm developed from Ref. [13]. The Hrinda optimization algorithm is used to test the arc length updating scheme. The finite-element program contained a 3-D truss element and an arc-length method that could solve the snap-through and snap-back response of structures presented by Crisfield in Refs. [6,7,10]. Examples of highly nonlinear trusses that have been found in the literature are presented to verify the method. Only conservative systems that adhere to the theorem of minimum potential energy are considered. These are stable systems that are subjected to quasi-static proportional loads, where the equations of equilibrium make the potential energy stationary under small displacements [14].

2. Arc length method

The Riks–Wempner arc-length method is used in this work to solve highly geometric nonlinear structures that are to be weight optimized. Various versions of the arc length method have been presented by many researchers [6,7,10,15,16] and incorporated into the finite-element analysis procedures. For convenience to the readers and to facilitate the discussion in subsequent sections, the Riks–Wempner arc length method is reviewed in Appendix A. The Riks–Wempner method is considered a partial arc length method because it relies on a normal to the tangent rather than the circular path to search for an equilibrium point [15]. The details of the iterative process along the normal are shown in Fig. A4 along with accompanying equations. The nonlinear equilibrium path is incrementally produced with load steps and convergence tests as depicted in Fig. A2.

3. Geometrically nonlinear 3-D truss element stiffness

In this work, we are interested in the behavior of 3-D truss members undergoing large displacements. In order to use and test the Hrinda optimization scheme, it became necessary to verify a 3-D truss stiffness matrix that could behave properly while undergoing large deflections. This requires a modification to the elastic stiffness that is typically used in the linear equation. The development of a properly formed geometric nonlinear stiffness matrix begins by altering the basic linear equation of the applied force F with the stiffness $[K]$ and nodal displacements u which can be written as

$$F = [K]u \tag{3.1}$$

The relationship is modified by introducing a nonlinear geometric matrix $[K]_G$ that is added to the elastic stiffness $[K]_E$ and written as

$$[K] = [K]_E + [K]_G \tag{3.2}$$

The resulting matrix is called the tangent stiffness and is written as

$$K_T = \frac{EA}{xl}\left\{\begin{array}{l} 1+\dfrac{[3(ux_2-ux_1)]}{xl}+\dfrac{[3(uy_2-uy_1)]}{xl}+\dfrac{[3(uz_2-uz_1)]}{xl} \\ \quad +\dfrac{[3(ux_2-ux_1)^2]}{2xl^2}+\dfrac{[3(uy_2-uy_1)^2]}{2xl^2}+\dfrac{[3(uz_2-uz_1)^2]}{2xl^2} \end{array}\right\} \tag{3.3}$$

Appendix B offers a detailed formulation of the tangent stiffness matrix given by Eq. (3.3) and the testing that was required to assure it was working properly.

4. Derivations of optimum nonlinear SED formulas

The discussion of using the SED as an indicator of an optimal design begins with an explanation of the strain and potential energy in truss elements. The optimization problem can then be defined with a recurrence formula that is developed to update the design variable, in this case, the cross-sectional area of the member.

Green's strain is used to develop the SED relationship in the 3-D truss element. This is the same strain measurement used in developing the tangent stiffness derived in Appendix B and shown in Eq. (3.3). A truss element's mass density is included in the strain equation to obtain the level of strain energy in the element. Appendix C.1 explains the derivation of the SED and shows the relationship in Eq. (C.8).

This relationship is used in developing the recurrence formula that modifies the cross-sectional area to reach a minimal weight design. Obtaining an updating recurrence relationship is possible by formulating a nonlinear constrained optimization problem that uses the potential energy as an equality constraint. The first step to solving the nonlinear constrained optimal problem and deriving an update scheme for the area is to form the Lagrangian. The method will answer the question "How do I minimize the SED while making sure the structure will support a load without becoming unstable?" The method is the basic tool used in nonlinear constrained optimization problems. Appendix C.2 goes through the derivation of the optimality criterion using the potential energy as an equality constraint and how it is incorporated into the Lagrangian. The discussion proceeds with differentiating with respect to the design variable to obtain the minimum potential energy. The SED is introduced to obtain a Lagrande multiplier γ given in Eq. (C.36) and reproduced below:

$$\gamma_i = \frac{\sum_{i=1}^{n} SED}{\sum_{i=1}^{n} SED^2} \tag{4.1}$$

The Lagrande multiplier is placed into the recurrence formula (C.35) to obtain the final form of the recurrence formula given by Eq. (C.37) and also shown below:

$$(A_i)_{\text{iter}+1} = (A_i)_{\text{iter}} \left(\frac{\sum_{i=1}^{n} SED}{\sum_{i=1}^{n} SED^2} \right) (SED_i)^{\lambda} \tag{4.2}$$

This equation is the recurrence relation that is used to update the design variable A_i at the end of each converged arc length iteration. This difference equation defines each term of the sequence as a function of the preceding terms. The equation gives a new A_i, based on the SED differences in all of the members. As an optimal solution is approached, the strain energy densities will become equal within an allowable user-specified difference.

5. Design-variable scaling

The arc length method is used in the Hrinda optimization algorithm to both solve the nonlinear structural system and also as an update indicator to monitor the SED in the truss elements of the system. As the equilibrium path is defined, parameters of the arc length are monitored and used to update the recurrence equation. The approach allows for updating the truss stiffness after each run with the arc length solver. As a critical-limit point is approached, the slope of the tangent to the equilibrium path approaches zero. The ideal solution would place the critical-limit load at the point which the slope goes to zero and all of the member strain energy densities become equal.

A recurrence relation is used to update the design variables in the Hrinda optimization algorithm, in this case, the cross-sectional areas of the truss members. Recurrence equations are used in iterative processes to keep track of and adjust variables in optimizations. Formulation of the updating recurrence relation relies on information from the arc length parameters. One of those parameters is the incremental external-load multiplier. The multiplier that is selected to update the recurrence equation depends on which phase the arc length solver is being processed. The solver has three phases; each phase checks the difference between the present total external load multiplier and the next load multiplier. This process is shown in the Hrinda optimization algorithm flowchart of Fig. 2. The checks indicate how close the load is to reaching a limit point and whether the arc length is becoming more horizontal with a decreasing slope.

A new parameter used in the Hrinda optimization algorithm, called the scale strain parameter, is developed to use the size of the incremental external load at the end of each arc length iteration. The value of the scale strain parameter depends on the recurrence checks that are performed (see Fig. 2). The three possible values of the scale strain parameter SSP are

$$SSP = \lambda_0 + \Delta\lambda_0, \quad \lambda_i + \Delta\lambda_i \quad \text{or} \quad \lambda_k + \Delta\lambda_k \tag{5.1}$$

where $\Delta\lambda_0$, $\Delta\lambda_i$ and $\Delta\lambda_k$ are the incremental load multipliers that are used along the load path. The initial load increment is $\Delta\lambda_0 = dl0$ and is supplied by the user at the start of the arc length procedure, as shown in Fig. 1. The value λ_0 is a preload value that is placed on the structure; this value is zero for the problems shown in this document. Fig. 1 shows the other two incremental multipliers as $\Delta\lambda_i = dli$, which is the incremental external multiplier at step i and $\Delta\lambda_k = dlk$, which is the incremental external multiplier at step k.

The new recurrence formula for scaling the design variable is taken from Eq. (C.37) and can be written by using the SSP as

$$SEDscale = [(SEDtotal/SEDtotal2)SED(i)]^{SSP} \tag{5.2}$$

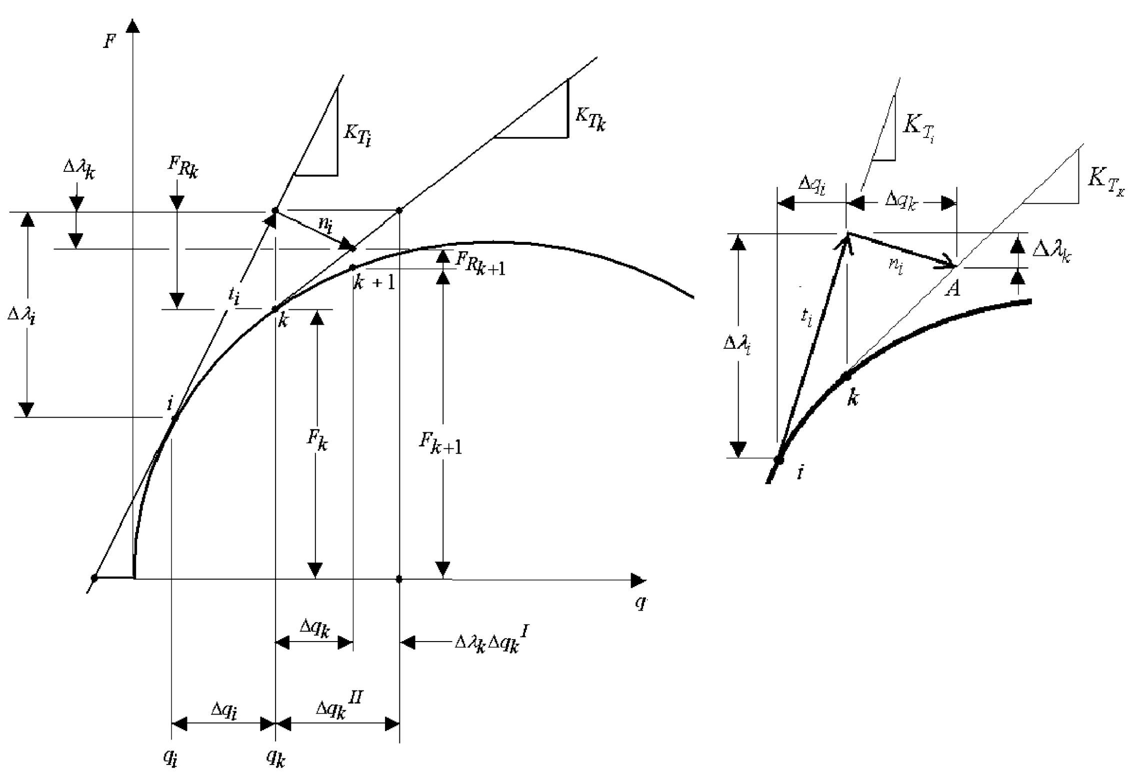

Fig. 1. The Riks–Wempner arc length method on a normal plane for a single-degree-of-freedom system.

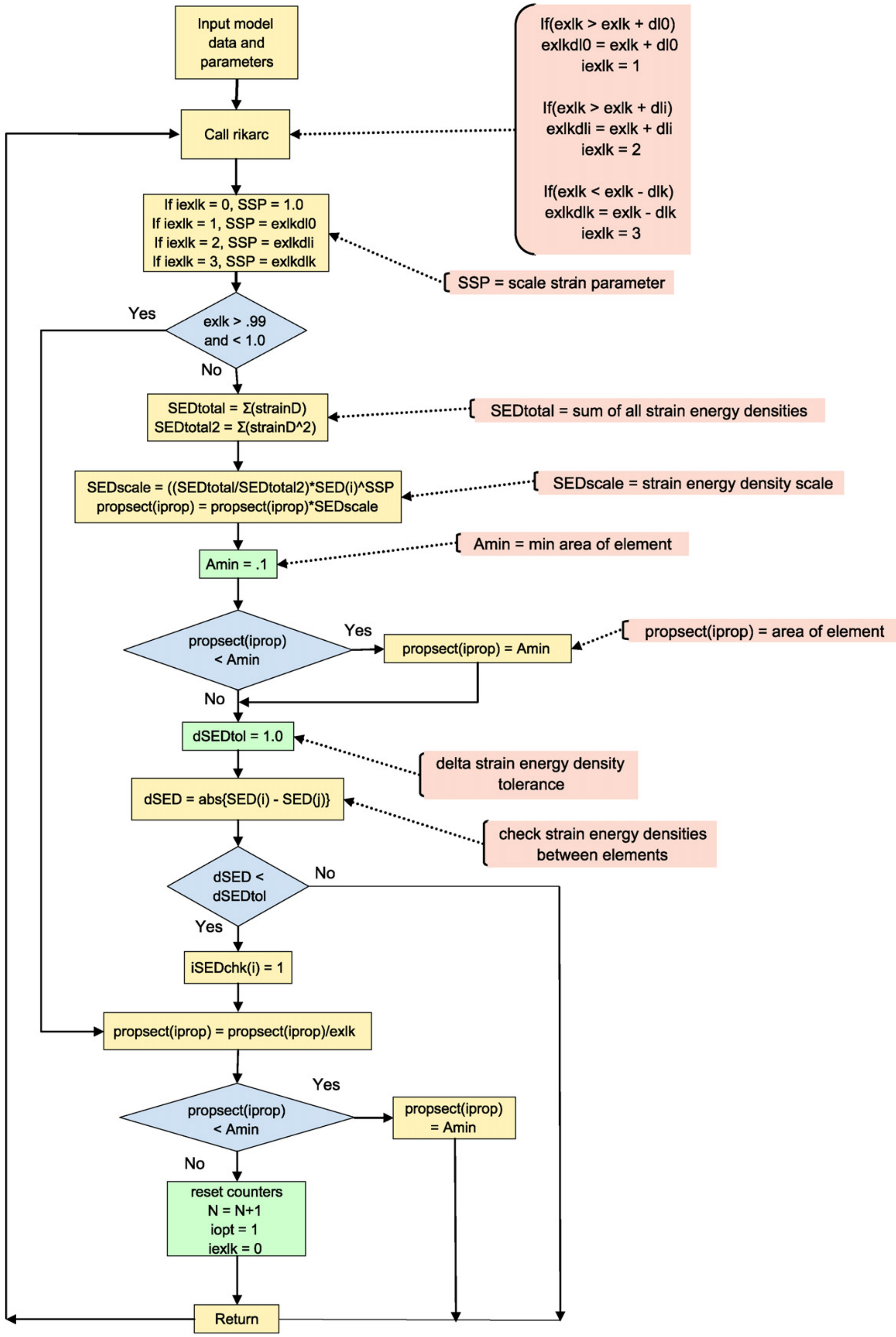

Fig. 2. Flowchart for the Hrinda optimizer program.

where

$SEDscale$ is the strain energy density scaling factor (5.3)

$SEDtotal$ is the total strain energy density (5.4)

$SEDtotal2$ is $SEDtotal^2$ (5.5)

$SED(i)$ is the strain energy density in the ith member (5.6)

SSP is the new scale strain parameter (5.7)

$dSEDtol =$ user-given differences between the strain energy densities for all elements (5.8)

The flowchart that is shown in Fig. 2 outlines the scheme for updating the area design variable *propsect*. Immediately after a solution is found in the arc length solver, the elemental strain energies are calculated and compared with the SED tolerance *dSEDtol*. If all members are within the tolerance, then the solution is optimal. If one member fails the tolerance test, then a new *SSP* is calculated from the initial multiplier *dl*0 or the incremental external multipliers *dli* and *dlk*.

The SED check is shown in the flowchart in Fig. 2 and graphically explained in Fig. 1. Each user-defined tolerance depends on the type of model, the level of fidelity that the user selects for in weight reduction, and how the structure was initially modeled. The whole structure can be optimized for weight, or truss members can be grouped together for sizing. If the absolute difference in SED is below the tolerance, then the solution is considered to be at its optimum. A minimum area is used to avoid zero-area trusses. A zero-area indicates that the truss member is necessary for a stable structure but carries no load. The minimum-area check is made to ensure that member sizes are greater than the user-provided area.

Another check of all of the strain energy densities is made using the user-provided *dSEDtol*. Only solid rods are used in this work, but any section could be used. The new strain energy densities are then computed with the updated areas.

The strategies that are explained for using the recurrence formula will be demonstrated using sample problems that are solved using a nonlinear solution in Nastran. The results of other nonlinear structures that have been published in numerous references will also be used to verify the Hrinda algorithm with the SED and arc length procedure.

6. Verification examples of optimized trusses

The following sample problems are used to verify the proposed nonlinear optimization method. All of the examples attempt to find the minimum weight of truss areas to support an applied load without buckling. The problems are shallow trusses that have critical-limit points and are taken from Khot and Kamat [9]. The first two problems solve a two-element, two-dimensional (2-D) system.

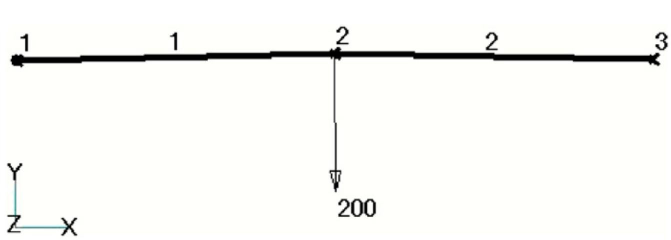

Fig. 3. Two-element symmetric truss with 200-lb apex load.

Iteration	Weight, lb
1	0.100020E+03
2	0.803313E+02
3	0.799023E+02
4	0.162505E+03

Fig. 5. Iteration weight history for two-element symmetric truss.

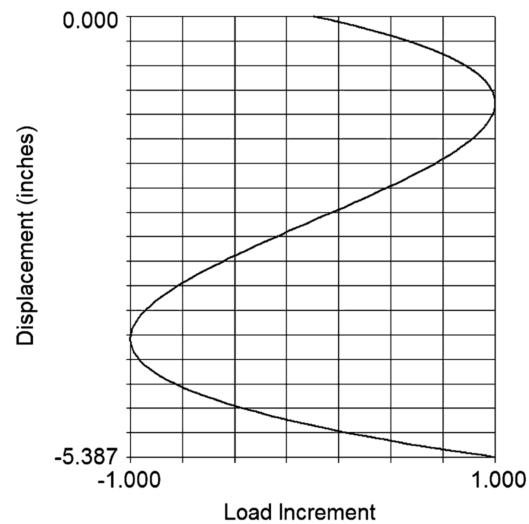

Fig. 6. Khot and Kamat load displacement curve for two-member symmetric truss with cross-section areas equal to 6.4977 in^2.

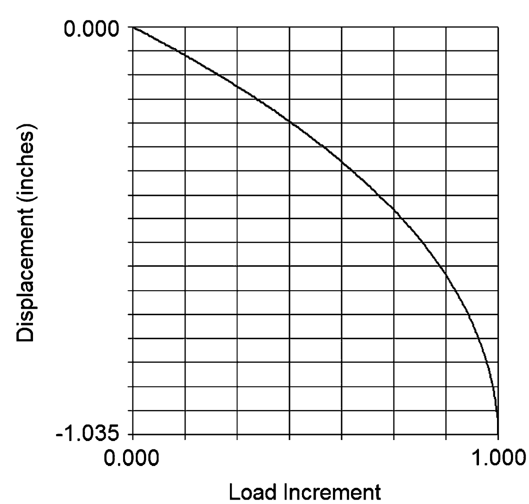

Fig. 7. Hrinda load displacement curve for two-member symmetric truss with cross-section areas equal to 6.4999 in^2.

Element	Initial Area, in^2	Initial SED	Initial Relative SED	Final SED	Final Relative SED	Final Area, in^2
1	3.00	1.38806	1.00000	0.88199	1.00000	6.4989
2	5.00	0.49960	0.35993	0.88199	1.00000	6.4989

Fig. 4. Strain energy density distribution for two-element symmetric truss.

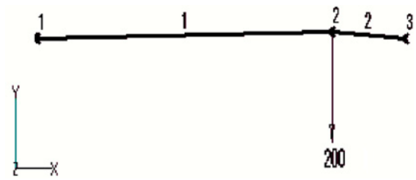

Fig. 8. Two-element asymmetric truss.

Element	Initial SED	Initial Relative SED	Final SED	Final Relative SED
1	1.50354	0.24968	2.15244	0.99961
2	6.02195	1.00000	2.15159	1.00000

Fig. 9. Strain energy density distribution of two-element asymmetric truss.

Iteration	Weight, (lb)
1	0.4501E+02
2	0.2414E+02
3	0.2461E+02
4	0.2483E+02
5	0.2490E+02
6	0.2492E+02
7	0.2493E+02
8	0.6664E+02

Fig. 10. Iteration weight history of two-element asymmetric truss.

A vertical load is applied at the mid-point with pinned end conditions. The next problem is a four-element system that has four different lengths with four cross sections to be optimized. Then, a larger star-dome problem is presented, which is similar to the Crisfield star dome in volume 2 [7]. This problem was minimized by using four cross-sectional area groups that were related to the layout of the design. The last problem is a large shallow truss that is more correct to space trusses. The problem allows for each of the members to be solved for a unique design variable. Here, the system has 13 cross sections that are sized to support a vertical load without snapping through. The solutions that were found in all the examples were compared with those solutions that were published by Khot as well as with Nastran. The areas found by the proposed technique were used in the Nastran models. Here, the displacements of the Nastran solutions were checked against those found by the proposed algorithm. The first example results are also compared with an analytical solution given by Crisfield [6].

6.1. Example 1. Two-member symmetric truss

The first example that is used to verify the Hrinda algorithm is a two-member symmetric shallow truss with a 200-lb concentrated load, as shown in Fig. 3. The problem is taken from Khot and Kamat [9] and is optimized to find the minimum sectional area that can support the 200-lb load without a sudden strain energy release and structure snap-through. All degrees of freedom were constrained except at the center node.

Two different initial areas of 3.0 in^2 and 5.0 in^2 were chosen to test the ability of the algorithms to find a unique area for each member. Fig. 4 shows the initial and final SED distributions in the two elements. The SED in the members become equal at 0.88199, which makes the final relative SED equal to 1.0 to indicate an optimal design. Both members were sized to equal areas of 6.499 in^2 which was slightly larger than the results that were found by Khot and Kamat in [9] (i.e. 6.4977 in^2 for both members). The total weight summary of the members after each iteration is given in Fig. 5.

The published results given in Ref. [9] used an area of 6.4977 in^2 and are close to optimal. However, the Nastran and analytical results as well as the Hrinda solution show the truss will not support the 200-lb load; snap-through buckling will still occur. The smaller area (i.e. 6.4977 in^2 from [9] was used in a Nastran analysis to obtain the results that are plotted in Fig. 6. The equilibrium path shows two critical-limit points, with the first at a load increment of 0.99959, or an applied load of 200 lb times 0.99959, which equals 199.9180 lb. The first limit point is reached, and then the structure responds with a snap-through before reaching the intended design load. An analytical solution was also obtained using the exact load displacement path equation given in Crisfield [6]. Here the analytical result also showed that a slightly larger area of 6.499 in^2 was required to support the 200-lb load.

The areas obtained from the Hrinda algorithm were substituted into the Nastran model of the two-element symmetric truss that was used in Fig. 3. The areas for both elements were the same as can be expected from symmetry (i.e. 6.4999 in^2). This optimal cross section produces a maximum displacement of 1.0354 in at the maximum load increment of 1.0, as shown in Fig. 7. The area obtained from the Hrinda algorithm was slightly higher than in [9] and did not cause an instability snap-through. The differences between the Hrinda and Khot/Kamat results are negligible. The two results demonstrate the sensitivity of the optimal solution to the number of significant digits used in the design and solution parameters. If a larger load increment is used during the arc length solution, then the design area may be slightly smaller, as in Ref. [9]. The two designs are nearly identical but illustrate how a slight imperfection can affect the design, as stated by Bruns et al. [17], Khot and Kamat [9] and Sedaghati et al. [12].

6.2. Example 2. Two-member asymmetric truss

The next verification problem is an asymmetric, two-element truss, shown in Fig. 8. The two-element truss has the same 200-lb applied load and support conditions as in the first example. The problem is also taken from Khot and Kamat [9]. A summary of the strain element density results that were obtained with the Hrinda algorithm is given in Fig. 9. The optimal area is achieved after the difference between the SED in both members falls within the user-defined parameter *dSEDtol*. The final optimal areas were6658 and 2.6646 in^2; for a practical design, they were the same in both members. The weight history is shown in Fig. 10 for each iteration, with a final design weight of 66.64 lb. This result compares closely with the result from Khot and Kamat of 66.53 lb [9].

6.3. Example 3. Four-member asymmetric truss

The next problem that was investigated was an optimized, asymmetric four-element truss with stability constraints that were presented by Khot and Kamat [9] and analyzed using the Hrinda optimization approach. The model is shown in Fig. 11 with a vertical loading of 200 lb at the center node. The first part of the analysis was to create a nonlinear Nastran model and test the results that were given in Ref. [9] for possible snap-through behavior. The areas that were used in the Nastran analysis were the final design areas from Ref. [9]; these are shown in Fig. 13. The load deflection plot of the center node is displayed in Fig. 15; the plot shows that a critical-limit point is reached before the design load of 200 lb. A critical-limit point is reached at a load increment of 0.9174 with a displacement equal to 1.2597 in. Strain energy that is stored in the structure is released, and the load decreases with increasing

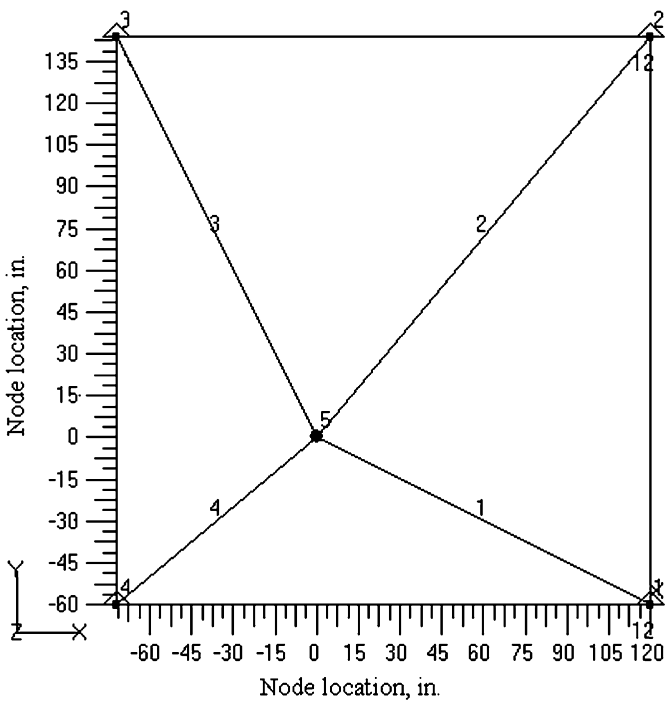

Fig. 11. Plan view of the four-member asymmetric truss example.

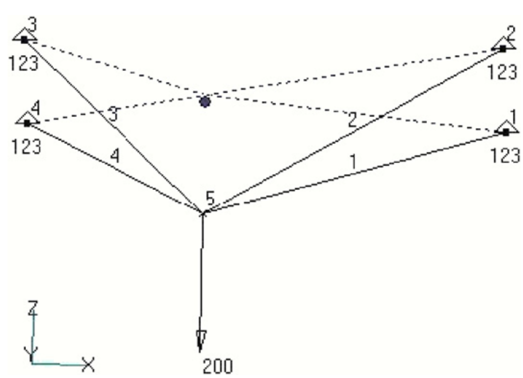

Fig. 12. Four-member truss at snap-through.

Element	Area, in^2
1	2.1555
2	1.6668
3	1.7202
4	2.0894

Fig. 13. Nastran element properties from Ref. [9].

Element	Area, in^2
1	2.1613
2	1.6596
3	1.7269
4	2.8940

Fig. 14. Hrinda final design.

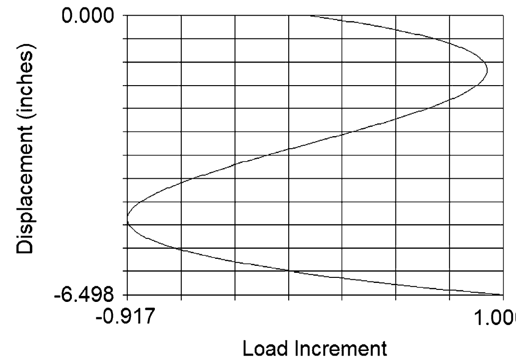

Fig. 15. Four-member asymmetric truss equilibrium path during snap-through.

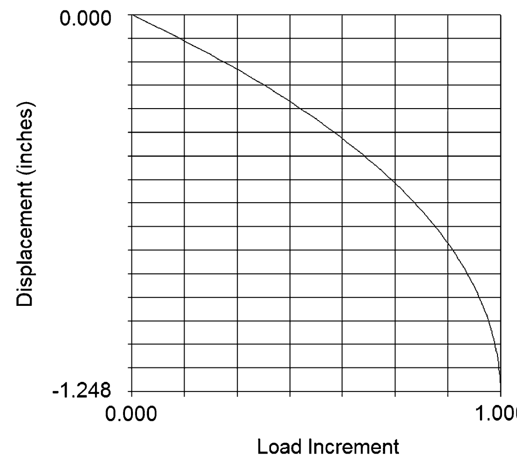

Fig. 16. Load–displacement results for center node of four-member truss using Hrinda results in Nastran.

load. The load goes to zero at about 3.0 in and changes direction until another limit point is reached at a maximum load increment of 0.9174 at a displacement equal to 4.7217 in. The curve shown in Fig. 15 demonstrates that the structure is only able to support a load of 0.9174×200 lb or 183.48 lb. Fig. 12 is a plot of the deflected shape after snap-through.

Now the Hrinda optimization algorithm method is used to solve and optimize the same problem that was given in Ref. [9]. The initial area for all of the members was 2.20 in^2, with the final areas shown in Fig. 14. The final optimized weight is 114.934 lb, as shown in the iteration history that is given in Fig. 18. The final results that were derived from the Hrinda method were used in a Nastran nonlinear analysis; the results for the center node are plotted in Fig. 16.

The load increment in Fig. 16 goes to 1.0 with a displacement of 1.248 in. This plot demonstrates that the structure can support the 200-lb load without snap-through buckling. The maximum load increment is reached at 1.0 and is multiplied by the applied load of 200 lb. The SED distribution results found by Hrinda are summarized in Fig. 17 and show that the strain energy densities in all of the members are nearly equal.

6.4. Example 4. Star-dome truss

The next problem that is presented is the shallow star-dome truss example that is introduced by Khot and Kamat [9] and shown in Fig. 19 . The design has 30 members with pin supports at the outer nodes and one vertical load at the center. The design sizing is formed on four optimization groups, which are listed in Fig. 20. The minimum area for all elements is 0.1 in^2.

Element	Initial Area, in^2	Initial SED	Initial Relative SED	Final Area, in^2	Final SED	Final Relative SED
1	2.20	0.105498E+01	0.558117	0.217240E+01	0.150140E+01	0.983680
2	2.20	0.620698E+00	0.328368	0.163832E+01	0.152631E+01	1.000000
3	2.20	0.673578E+00	0.356343	0.172547E+01	0.152479E+01	0.999004
4	2.20	0.189025E+01	1.000000	0.292330E+01	0.147108E+01	0.963815

Fig. 17. Strain energy density distribution in four-member truss.

Iteration	Weight, lb.
1	0.126822E+03
2	0.121775E+03
3	0.118580E+03
4	0.116549E+03
5	0.115251E+03
6	0.114934E+03

Fig. 18. Iteration weight history for four-member truss.

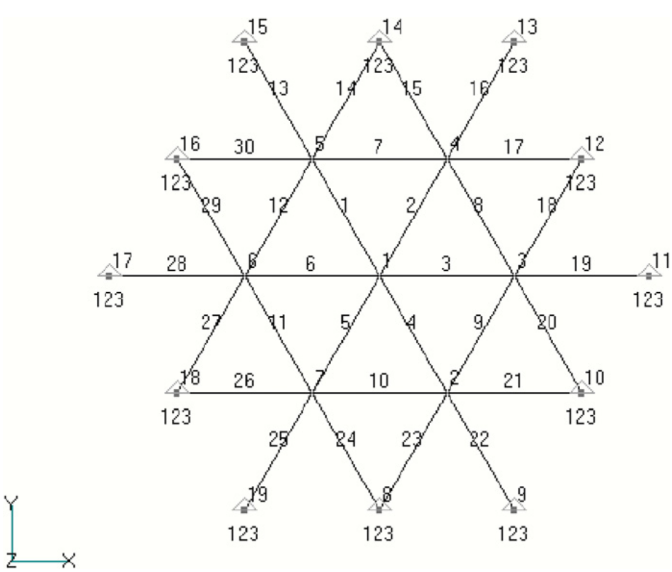

Fig. 19. Star-dome truss example.

Element	Initial SED	Initial Relative SED	Final SED	Final Relative SED	Final Area, in^2
1	19.428600	1.000000	15.493200	0.964167	1.69407
2	19.428600	1.000000	15.493200	0.964167	1.69407
3	19.428600	1.000000	15.493200	0.964167	1.69407
4	19.428600	1.000000	15.493200	0.964167	1.69407
5	19.428600	1.000000	15.493200	0.964167	1.69407
6	19.428600	1.000000	15.493200	0.964167	1.69407
7	12.605600	0.648817	15.475700	0.963078	1.37499
8	12.605600	0.648817	15.475700	0.963078	1.37499
9	12.605600	0.648817	15.475700	0.963078	1.37499
10	12.605600	0.648817	15.475700	0.963078	1.37499
11	12.605600	0.648817	15.475700	0.963078	1.37499
12	12.605600	0.648817	15.475700	0.963078	1.37499
13	0.740568	0.038117	16.069000	1.000000	0.263468
14	0.000002	0.000000	4.293670	0.267202	0.100379
15	0.000002	0.000000	4.293670	0.267202	0.100379
16	0.740568	0.038117	16.069000	1.000000	0.263468
17	0.000002	0.000000	4.293670	0.267202	0.100379
18	0.000002	0.000000	4.293680	0.267203	0.100379
19	0.740566	0.038117	16.069000	1.000000	0.263468
20	0.000002	0.000000	4.293680	0.267203	0.100379
21	0.000002	0.000000	4.293670	0.267202	0.100379
22	0.740568	0.038117	16.069000	1.000000	0.263468
23	0.000002	0.000000	4.293670	0.267202	0.100379
24	0.000002	0.000000	4.293670	0.267202	0.100379
25	0.740568	0.038117	16.069000	1.000000	0.263468
26	0.000002	0.000000	4.293670	0.267202	0.100379
27	0.000002	0.000000	4.293680	0.267203	0.100379
28	0.740566	0.038117	16.069000	1.000000	0.263468
29	0.000002	0.000000	4.293680	0.267203	0.100379
30	0.000002	0.000000	4.293670	0.267202	0.100379

Fig. 21. Star-dome strain energy density and truss area design.

Group I	1 - 6
Group II	7 - 12
Group III	13, 16, 19, 22, 25, 28
Group IV	14, 15, 17, 18, 20, 21, 23, 24, 26, 27, 29, 30

Fig. 20. Star-dome optimization groups.

Iteration	Weight, lb
1	0.1087E+04
2	0.4963E+03
3	0.5165E+03
4	0.5520E+03
5	0.5913E+03
6	0.6384E+03
7	0.6942E+03
8	0.7622E+03
9 (final)	**0.7650E+03**

Fig. 22. Star-dome weight summary.

The SED summary and the final member areas are given in Fig. 21. The relative SED are also given; group IV is sized to the minimum $0.10\,in^2$. Group IV members have been sized to the minimum area so that their relative strain energy densities do not approach unity. Fig. 22 shows the total design weights after each iteration, with a final weight of 765.0 lb. This weight compares well with the final optimized design weight of 766.188 lb given by Khot and Kamat [9]. The group sizing results are shown in Fig. 23; these results also compare closely with the results from Khot and Kamat [9].

Group	Area, in^2 Hrinda	Area, in^2 Khot/Kamat
I	1.69407	1.6926
II	1.37499	1.3754
III	0.263468	0.2693
IV	0.100379	0.1000

Fig. 23. Final optimized group truss areas.

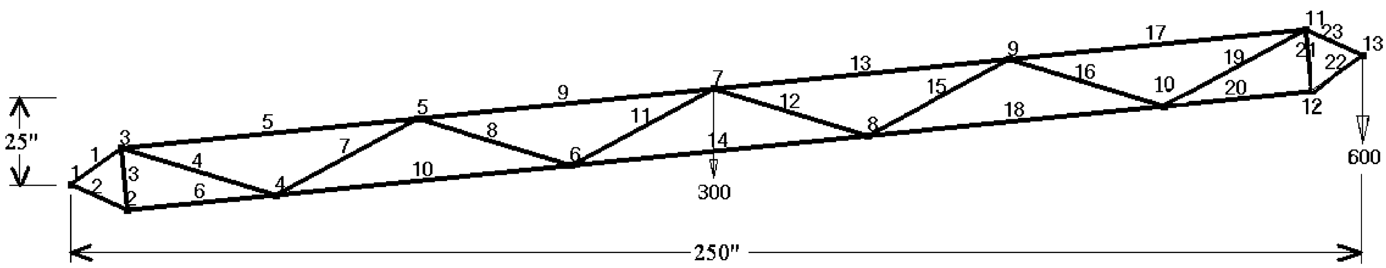

Fig. 24. Shallow truss half-symmetry model with two point loads.

Element	Initial SED	Initial Relative SED	Final SED	Final Relative SED	Final Area, in^2 (Hrinda)	Final Area, in^2 (Khot)
1	2.65769	2.65769	1.59397	0.99744	2.5794	2.4747
2	2.30213	2.30213	1.59178	0.99607	2.4089	2.3695
3	5.75174	0.57517	1.56639	0.98018	1.2110	1.1837
4	2.09883	0.02098	1.50676	0.94287	0.2307	0.2609
5	2.38905	2.38905	1.59195	0.99618	2.4434	2.3694
6	1.72627	1.72627	1.58652	0.99278	2.0888	2.0551
7	2.07244	0.02072	1.50816	0.94375	0.2306	0.2259
8	1.99700	0.01997	1.50975	0.94474	0.2292	0.2227
9	3.27302	3.27302	1.59805	1.00000	2.8546	2.7512
10	1.09653	1.09653	1.57834	0.98766	1.6750	1.6353
11	1.93285	0.01932	1.50944	0.94455	0.2264	0.1945
12	1.92360	0.01923	1.50927	0.94444	0.2258	0.1941
13	3.23698	3.23698	1.59784	0.99986	2.8390	2.7463
14	6.21472	0.62147	1.56710	0.98063	1.2654	1.2751
15	1.96688	0.01966	1.50947	0.94457	0.2274	0.2230
16	2.02073	0.02020	1.50774	0.94348	0.2277	0.2249
17	2.36631	2.36631	1.59177	0.99607	2.4319	2.3649
18	1.09379	1.09379	1.57828	0.98762	1.6728	1.6351
19	2.06643	0.02066	1.50650	0.94271	0.2289	0.2590
20	1.71562	1.71562	1.58640	0.99271	2.0824	0.0524
21	5.64525	0.56452	1.56604	0.97996	1.1998	1.1756
22	2.28075	2.28075	1.59160	0.99596	2.3978	2.3626
23	2.62850	2.62850	1.59375	0.99730	2.5653	2.4687

Fig. 25. Shallow truss strain energy density and final design areas.

Iteration	Weight, lb
1	0.156524E+03
2	0.134545E+03
3	0.123092E+03
4	0.116857E+03
5	0.113505E+03
6	0.111891E+03
7	0.111382E+03
8	0.111581E+03
9	0.112211E+03
10	0.113076E+03
11	0.114034E+03
12	0.114992E+03
13	0.115892E+03
14	0.116704E+03
15	0.117415E+03
16	0.118024E+03
17	0.118539E+03
18	0.118968E+03
19	0.119322E+03
20	0.119612E+03
21 (final)	**0.119849E+03**

Fig. 26. Shallow truss weight summary.

6.5. Example 5. Large shallow truss

The next problem is a large symmetric shallow truss, which is presented by Khot and Kamat in Ref. [9]. The design (see Fig. 24) includes 23 members that are pinned at node 1 and sized for two applied loads of 300 and 600 lb with symmetric boundary conditions at the apex.

The SED summary and the final member areas are shown in Fig. 25. A *dSEDtol* = 0.10 is imposed for all elements. The member areas that are derived with the Hrinda optimization algorithm are listed next to those of Khot and Kamat [9] for comparison. All of the members are in close agreement except member 20. This is because the Hrinda results imposed a minimum area of 0.1 in^2 so that the optimizer was forced to size accordingly. The Hrinda weight summary is given in Fig. 26 with the final weight of 119.85 lb. Doubling this value to account for symmetry yields the final shallow truss weight of 239.70 lb, which is close to the final weight of 234.13 lb found by Khot and Kamat [9].

7. Discussion

The Lagrange multiplier is written as a function of the strain energy densities in Eq. (C.36). The equation is placed into the design variable Eq. (C.35) to form a simple updating Eq. (C.37). The exponential recurrence formula (C.37) outperformed other updating relationships by being able to adjust the design variable to obtain the correct minimum weights given by the Section 6 examples. Other relationships were attempted; however, they could not achieve a uniform SED.

Several members of the "star-dome truss" in Example 4 were sized to the design minimal area of 0.1 in^2. Fig. 21 lists the member results and show members: 14, 15, 17, 18, 20, 21, 23, 24, 26, 27, 29 and 30 as approaching the design minimal area. This is due to the light loads in the members or members that have no load in them and are passive. The same condition also exists for member 20 of Example 5. Sizing results for member 20 are listed in Fig. 25 and show the member being sized to a minimal area indicating that it is very lightly loaded or has no load. This is typical for truss structures in which some members are passive during certain applied loads. The condition may change as the applied loading is placed at different nodes on the structure or if the load direction is changed.

Modern design of lightweight truss structures has been achieved through the use of analysis techniques that were developed to investigate nonlinear instabilities. The application of loads to a truss system that produces nonproportional displacements is nonlinear; increases in the loading will make the structure unstable or cause it to buckle. In Ref. [15], these systems are called highly flexible structures (HFS) and are designed for large elastic displacements and without any plastic deformation. These HFS may either show bifurcation or limit-point instability. Bifurcation instability occurs when the structure has a sudden geometric change and is characterized by an abrupt equilibrium path change along the load path. Strain energy from a compressive force is converted to bending strain energy without a large force increase. This conversion can best be visualized as a typical Euler buckling of a slender column. The second type of nonlinear structural response is limit-point instability. This behavior is represented as a gradually changing equilibrium path with no sudden changes. Increasing the load beyond the limit point causes a release of strain energy as the stiffness becomes negative.

The Hrinda optimization algorithm did check for local Euler buckling in the members. The program would perform a simple buckling check for a pinned–pinned slender column using the internal loads. The examples tested did not have any individual members buckle before reaching an instability point of the total structure. This could change if the initial geometry of the structure is altered enough to stiffen the structure and prevent a global instability.

Lightweight truss structures are often highly flexible with unpredictable nonlinear responses and may become unstable before a bifurcation point is reached. An understanding of elastic stability behavior in nonlinear structures can be gained by graphically representing the load and the displacement (i.e. the equilibrium path of the structure).

Methods that overcome the challenges of accurately identifying nonlinear equilibrium paths have been improving and also have been developed to include optimization at limit points. With varying success, growing number of attempts have been made to optimize a structure at a limit load. The challenge is to design a flexible structure to undergo large displacements and remain functional without a loss of stability.

8. Conclusions

The optimization of stability-constrained truss structures that exhibit highly nonlinear snap- through behavior has been accomplished using the arc length method and the principal of strain energy density. The iterative scheme for updating design variables was used in a finite-element approach that also used the critical buckling load at snap-through as the design load. The problems that were executed agreed closely with previous truss examples that were found in the literature. The results verify that the use of the arc length parameters can be used as performance indicators to update the design variables. The technique was applied to design problems with highly nonlinear trusses that are prone to global instability failures, but the technique has also been tested on linear optimization problems. Initial results show that the arc length and strain energy density parameters can be used to find the lightest truss that exhibits linear behavior.

Appendix A

A.1. Review of arc length formulas

The Riks–Wempner arc length method traces the nonlinear equilibrium path by using an iterative process that begins by computing initial displacements Δq_0 based on a user-defined load increment $\Delta\lambda_0$, as shown in Fig. A1. The linear stiffness K_{T_0} is used to start the process and is replaced with the tangent stiffness K_{T_i} in further iterations.

The initial displacements Δq_0 are found using

$$\frac{\Delta\lambda_0}{\Delta q_0} = \frac{\lambda}{\Delta q_{\mathrm{tot}}} \tag{A.1}$$

where $\lambda = 1$ and Δq_{tot} is derived from the expression

$$K_{T_0}\Delta q_{\mathrm{tot}} = \lambda Q \tag{A.2}$$

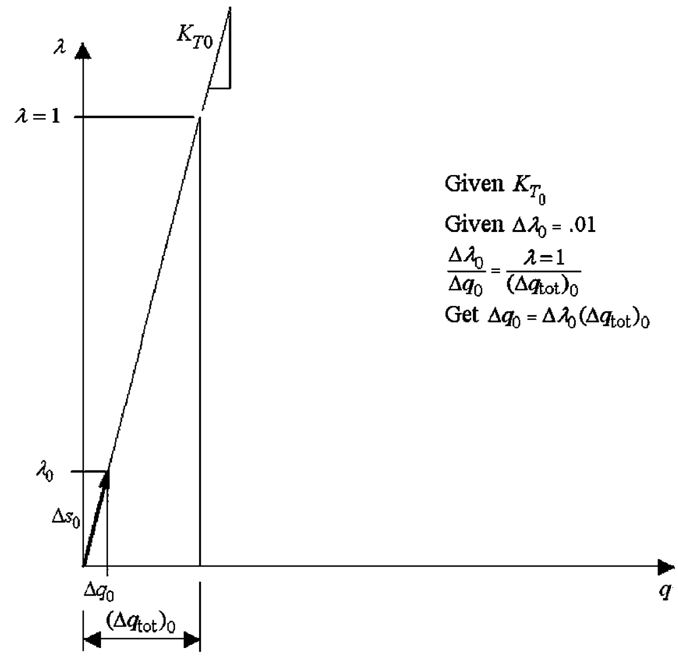

Fig. A1. Starting the arc length.

where Q is the total applied loading. The notations that are given here for the displacements and loads are vector quantities with the stiffness notation understood as a matrix. Substituting into Eq. (A.1) yields

$$\frac{\Delta\lambda_0}{\Delta q_0} = \frac{\lambda = 1}{(\Delta q_{\mathrm{tot}})_0} \tag{A.3}$$

which is solved for the initial displacement of

$$\Delta q_0 = \Delta\lambda_0(\Delta q_{\mathrm{tot}})_0 \tag{A.4}$$

The method proceeds to find the next equilibrium point from the initial point 0 by using the tangent stiffness K_{T_1} as shown in Fig. A2. The tangent stiffness matrix is assembled by using the nonlinear truss that is given in Eq. (B.28) in Appendix B; it is defined as the elastic linear stiffness that is given by Eq. (B.6) and added to the nonlinear stiffness in Eq. (B.26). The sum of the linear elastic and nonlinear matrices produces the global tangent stiffness at point i along the load–displacement path of the single-degree-of-freedom system. A new displacement $\Delta q_{\mathrm{tot}_i}$ is calculated from

$$K_{T_i}\Delta q_{\mathrm{tot}_i} = \lambda Q \tag{A.5}$$

with $\lambda = 1$ as before.

The arc length used in the Riks–Wempner method is the straight line Δs_i; this arc length is either constant or scaled by the user input with the following:

$$\Delta s_i = \Delta s_{i-1}\left(\frac{I_{\mathrm{DES}}}{I_{i-1}}\right)^{1/2} \tag{A.6}$$

The user decides on the required number of iterations I_{i-1} and on the number of desired iterations I_{des}. The arc length Δs_i is the tangent vector along the equilibrium path and can be calculated as

$$\Delta s_0 = \sqrt{\Delta\lambda_0^2 + (\Delta q^{\mathrm{T}})_0\Delta q^{\mathrm{T}}} \tag{A.7}$$

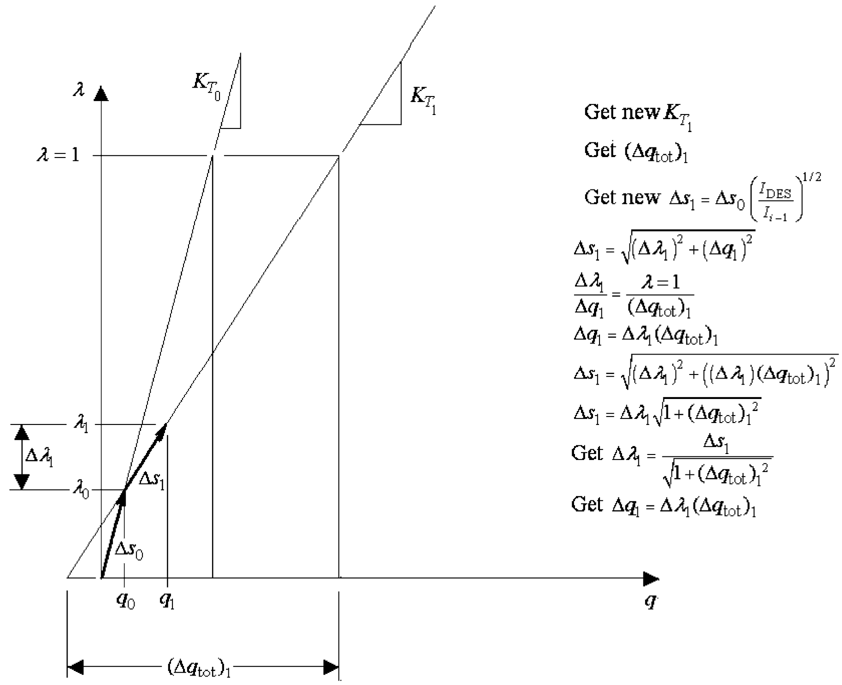

Fig. A2. Next iteration.

or by using

$$\Delta s_0 = \Delta\lambda_0 \sqrt{1 + \Delta q_{\text{tot}}^{\text{T}} \Delta q_{\text{tot}}} \tag{A.8}$$

The length of the next tangent vector Δs_1 can now be written as

$$\Delta s_1 = \Delta\lambda_1 \sqrt{1 + (\Delta q_{\text{tot}})_1^2} \tag{A.9}$$

In this equation, the unknown is $\Delta\lambda_1$, which can be solved by using

$$\Delta\lambda_1 = \frac{\Delta s_1}{\sqrt{1 + (\Delta q_{\text{tot}})_1^2}} \tag{A.10}$$

the incremental displacement can be found by using

$$\Delta q_1 = \Delta\lambda_1 (\Delta q_{\text{tot}})_1 \tag{A.11}$$

The next step is a test for convergence that first obtains the internal force F_1. The residual forces can be calculated as

$$F_{\text{R}_1} = \lambda_1 Q - F_1 \tag{A.12}$$

Here, the residual forces are the differences in the internal forces that are calculated from the equilibrium path and are shown in Fig. A3.

The following is used to test for load convergence:

$$\frac{F_{\text{R}_1}^{\text{T}}(s_1) F_{\text{R}_1}}{\lambda_1 \sqrt{Q^{\text{T}}(s_1) Q}} \leqslant \lambda_1 Q - \Delta f \tag{A.13}$$

where Δf is a user-defined tolerance value. If the statement is true, then the load test is successful.

A convergence for displacements is also performed by using the ratio

$$\left\| \frac{\Delta q_1}{q_1} \right\| \leqslant \delta_{\text{d}} \tag{A.14}$$

where δ_{d} is also a user-defined tolerance. Once again, if the statement is true, then the displacement test is passed.

If both convergence tests are passed, then another load step is started by following the same procedure that is outlined above. If, however, one of the tests fail, then iteration on the normal begins as shown in Fig. A4. The objective is to find q_2 and λ_2 by iterating down the normal from the tangent vector Δs_1. Fig. A4 shows the necessary geometry that is required to begin the iteration. A portion of the figure is magnified to show how the variables are defined.

The vector n_2 is normal to Δs_1 if

$$\frac{\Delta q_1}{\Delta\lambda_1} = \frac{\Delta\lambda_2}{\Delta q_2} \tag{A.15}$$

The equation is from using the method of similar triangles as shown in Fig. A5. The relationship may also be written as

$$\Delta q_1 \Delta q_2 - \Delta\lambda_1 \Delta\lambda_2 = 0 \tag{A.16}$$

A new tangent stiffness $[K_{\text{T}_2}]$ is calculated and used in the following equation:

$$[K_{\text{T}_2}] \Delta q_2^{\text{II}} = F_{\text{R}_1} \tag{A.17}$$

where all variables are defined in Fig. A5.

Eq. (A.17) can be rewritten as

$$\Delta q_2^{\text{II}} = [K_{\text{T}_2}]^{-1} F_{\text{R}_1} \tag{A.18}$$

The incremental displacement shown in Fig. A5 can be expressed as

$$\Delta q_2 = \Delta q_2^{\text{II}} - \Delta q_2^{*} \tag{A.19}$$

Using similar triangles, the following can now be obtained:

$$\frac{\Delta\lambda_2}{\Delta q_2^{*}} = \frac{(\lambda = 1)}{\Delta q_2^{\text{I}}} \tag{A.20}$$

Terms can be rearranged to obtain

$$\Delta q_2^{*} = \frac{\Delta\lambda_2 \Delta q_2^{\text{I}}}{(\lambda = 1)} = \Delta\lambda_2 \Delta q_2^{\text{I}} \tag{A.21}$$

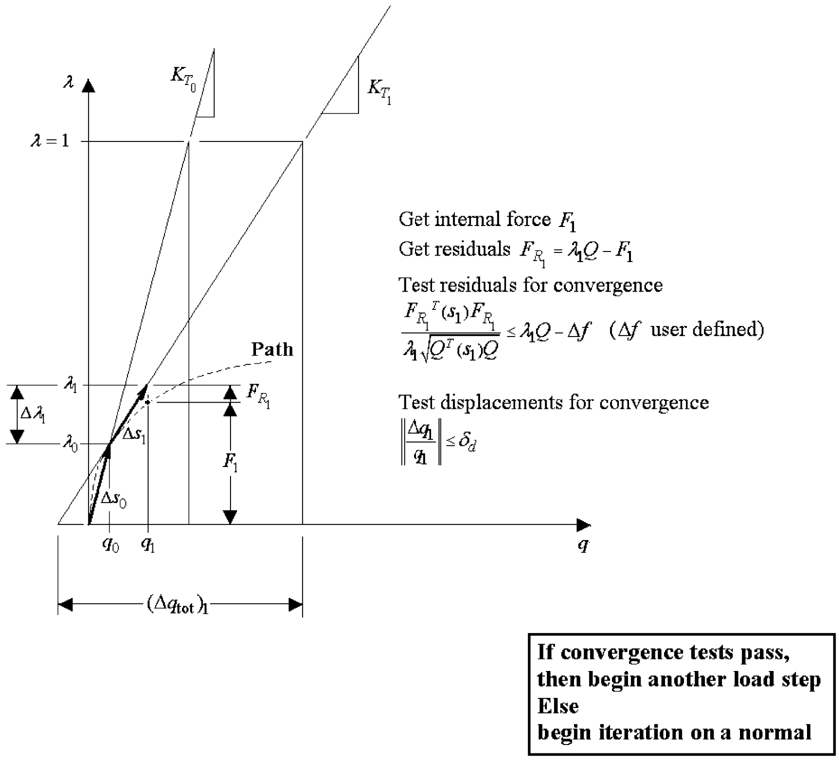

Fig. A3. Convergence tests.

Combining Eqs. (A.19) and (A.21) creates

$$\Delta q_2 = \Delta q_2^{\mathrm{II}} - \Delta\lambda_2 \Delta q_2^{\mathrm{I}} \tag{A.22}$$

Now, Eq. (A.16) is combined with Eq. (A.22) to obtain

$$\Delta q_1(\Delta q_2^{\mathrm{II}} - \Delta\lambda_2 \Delta q_2^{\mathrm{I}}) - \Delta\lambda_1 \Delta\lambda_2 = 0$$

which can be reduced to

$$\Delta\lambda_2 = \frac{\Delta q_1 \Delta q_2^{\mathrm{II}}}{(\Delta q_1 \Delta q_2^{\mathrm{I}} + \Delta\lambda_1)} \tag{A.23}$$

and added to the previous load increments as follows:

$$\lambda_2 = \lambda_0 + \Delta\lambda_1 + \Delta\lambda_2 \tag{A.24}$$

Likewise, the displacement increments are summed by using Eq. (A.22) to yield

$$q_2 = q_0 + \Delta q_1 + \Delta q_2 \tag{A.25}$$

Eqs. (A.24) and (A.25) represent the load and displacement at location 2 along the equilibrium path. A convergence check is performed similar to that shown in Fig. A3 and in Eqs. (A.12)–(A.14). If the test is passed, then a new load increment begins by using the tangent vector of Eq. (A.9). If the tests fail, then iteration down the normal vector n is started as shown in Fig. A6. The process continues until the convergence tests for load and displacement are both passed.

Structured state-of-the-art sparse equation solver technology such as sparse assembly algorithms, reordering algorithms to minimize fill-in-terms, and super node strategies for efficient unrolling techniques during numerical factorization phase were all incorporated into Eqs. (A.5) and (A.18) of the arc length procedure [18].

Appendix B

B.1. Derivation of 3-D geometrically nonlinear truss element stiffness

Trusses that are undergoing large deflections must be analyzed for the deformed geometry of the structure. The linear equation

$$F = [K]u \tag{B.1}$$

which relates the applied force F with the truss element stiffness $[K]$ and displacements u must be modified to account for changes in nodal geometry as the load is applied. The standard stiffness elastic stiffness matrix $[K]_{\mathrm{E}}$ is modified by the addition of a geometric stiffness matrix $[K]_{\mathrm{G}}$ that depends on the geometry and the initial internal forces. The elastic and geometric matrices can be written as

$$[K] = [K]_{\mathrm{E}} + [K]_{\mathrm{G}} \tag{B.2}$$

Note here that efficient sparse assembly procedures [18] have been employed in Eq. (B.2), while efficient sparse reordering algorithms (i.e. both to minimize fill-in terms and during the factorization phase) and sparse solvers (with unrolling strategies [18] have been utilized in Eq. (B.1).

Nonlinear analysis is usually divided into a sequence of load steps where the displacements are being solved after each load increment. The nonlinear stiffness is then updated with the displacement and the internal force values from the previous step. A discussion of the nonlinear truss element that is used in the finite-element program in this study follows.

The following definitions are used to derive the elastic truss stiffness with the variables that are defined in Fig. B1. The change in displacements for an ith truss can be written as

$$\Delta X^{(i)} = x_2 - x_1 + \Delta x_2 - \Delta x_1 \tag{B.3}$$

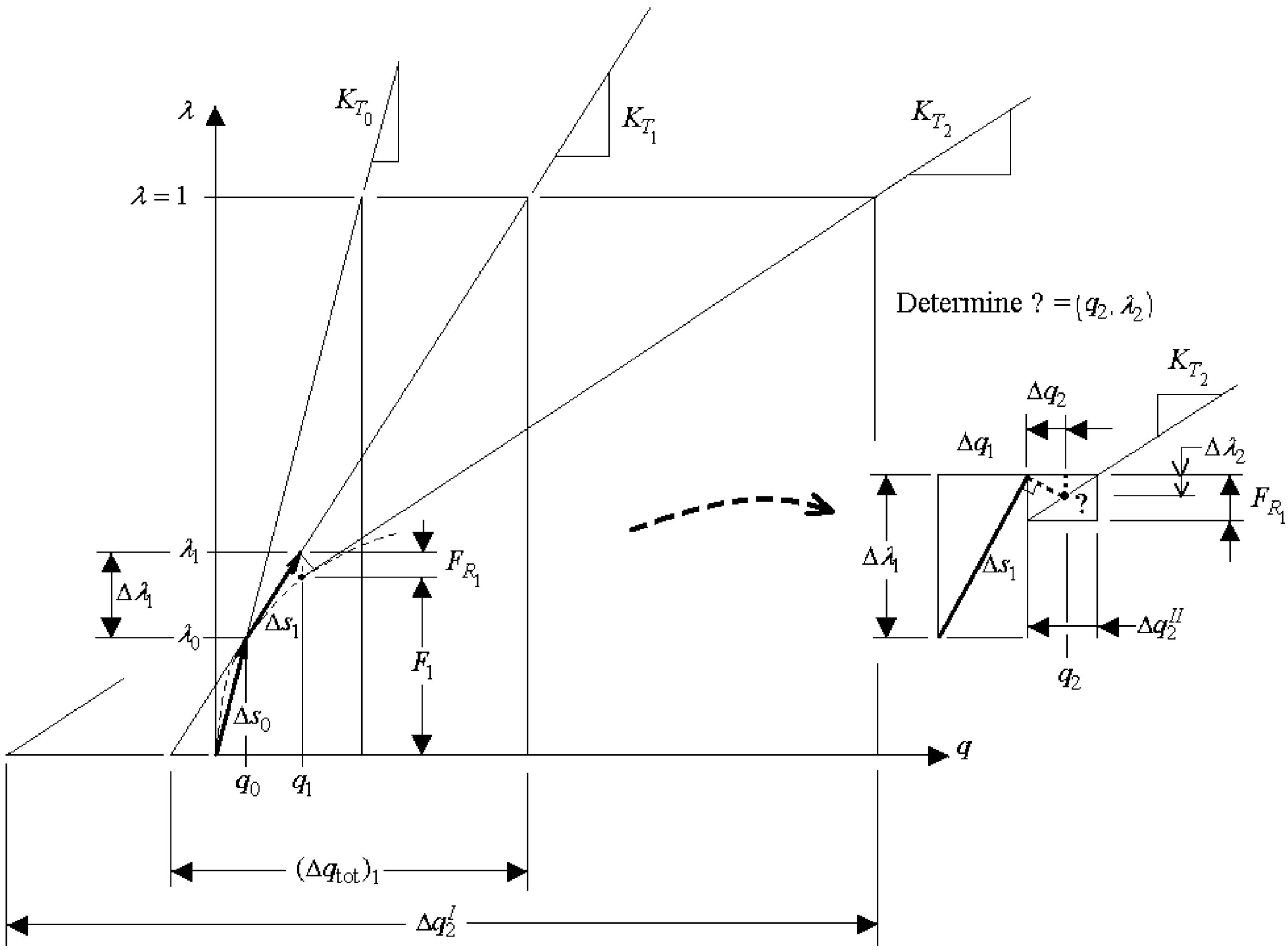

Fig. A4. Iteration on a normal.

$$\Delta Y^{(i)} = y_2 - y_1 + \Delta y_2 - \Delta y_1 \tag{B.4}$$

$$\Delta Z^{(i)} = z_2 - z_1 + \Delta z_2 - \Delta z_1 \tag{B.5}$$

where Δx_1, Δy_1, Δz_1 are the displacements at end 1 and Δx_2, Δy_2, Δz_2 are the displacements at end 2. The global nodal coordinates are given as x_1, y_1, z_1 at end 1 and x_2, y_2, z_2 for end 2. The original length is $L^{(i)}$ and the new length is given as $\Delta L^{(i)}$.

The matrix $[K]_E$ that is shown below is the normal elastic stiffness matrix for a 3-D truss defined with the use of variables for easier computer programming.

$$[K]_E = CON1\begin{bmatrix} C(1)C(1) & C(1)C(2) & C(1)C(3) & C(1)C(4) & C(1)C(5) & C(1)C(6) \\ & C(2)C(2) & C(2)C(3) & C(2)C(4) & C(2)C(5) & C(2)C(6) \\ & & C(3)C(3) & C(3)C(4) & C(3)C(5) & C(3)C(6) \\ & & & C(4)C(4) & C(4)C(5) & C(4)C(6) \\ & & & & C(5)C(5) & C(5)C(6) \\ & & & & & C(6)C(6) \end{bmatrix} \tag{B.6}$$

where the following are defined as

$$X21D = \Delta x_i \tag{B.7}$$

$$Y21D = \Delta y_i \tag{B.8}$$

$$Z21D = \Delta z_i \tag{B.9}$$

$$C(1) = -x21D \tag{B.10}$$

$$C(2) = -Y21D \tag{B.11}$$

$$C(3) = -Z21D \tag{B.12}$$

$$C(4) = -C(1) \tag{B.13}$$

$$C(5) = -C(2) \tag{B.14}$$

$$C(6) = -C(3) \tag{B.15}$$

$$CON1 = \frac{EA}{xl^3} \tag{B.16}$$

where EA is Young's modulus multiplied with the bar cross section. The nonlinear geometric stiffness is now defined and added to the elastic stiffness (Eq. (B.6)).

B.2. Element internal force

The internal forces in the truss element are required for the geometric stiffness and are defined by using matrix notation. The element strain formulation uses a constant cross-sectional area and assumes that the length and area of the truss will remain large. The strain energy or the work done is one-half of the nodal forces multiplied by the corresponding deflections.

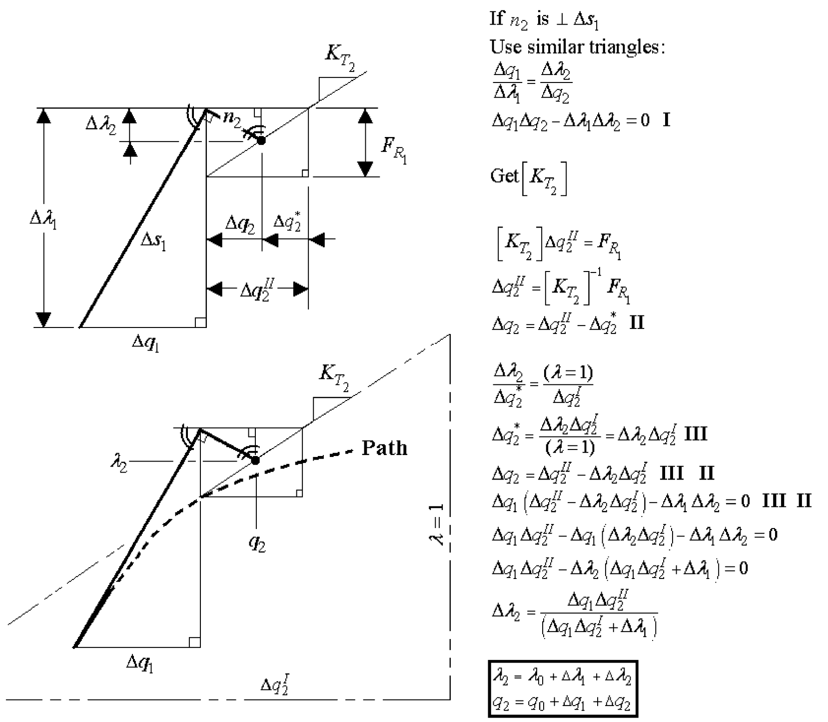

Fig. A5. Details of iteration on a normal.

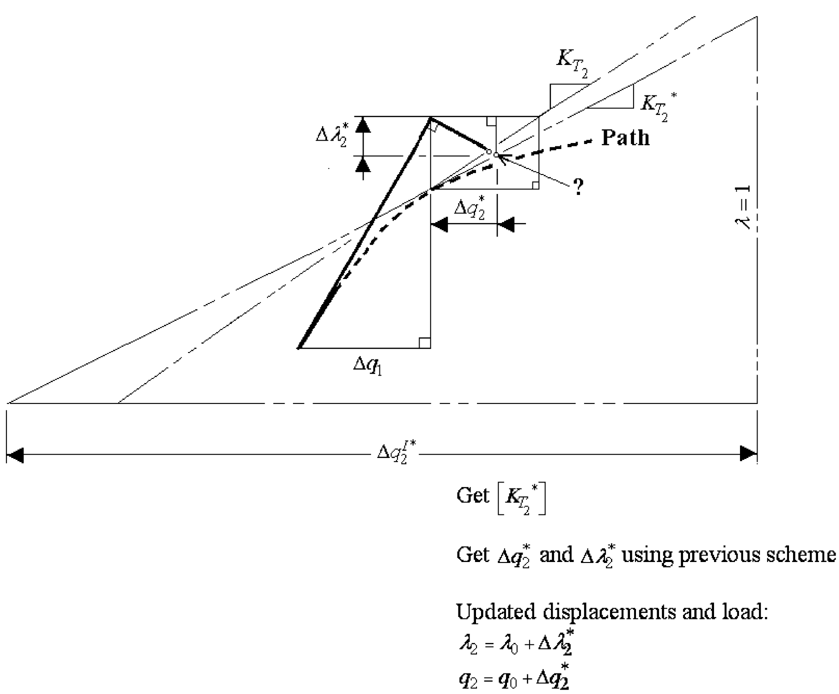

Fig. A6. Continuing iteration on normal.

The internal force in the truss is now defined to include nonlinear effects. The force will be axial and is needed to update the nonlinear stiffness matrix. Defining the force in matrix notation is necessary for efficient computer programming. The truss length from incremental displacements may be written as

$$xli^2 = (\vec{x}_{21} + \vec{p}_{21})^{\mathrm{T}} (\vec{x}_{21} + \vec{p}_{21}) \tag{B.17}$$

where

$$\vec{p}_{21} = \begin{bmatrix} ux_{21} \\ uy_{21} \\ uz_{21} \end{bmatrix} \quad \text{and} \quad \vec{xl}_{21} = \begin{bmatrix} (x_2 - x_1) \\ (y_2 - y_1) \\ (z_2 - z_1) \end{bmatrix} \tag{B.18}$$

with

$$ux_{21} = ux_2 - ux_1 \tag{B.19}$$

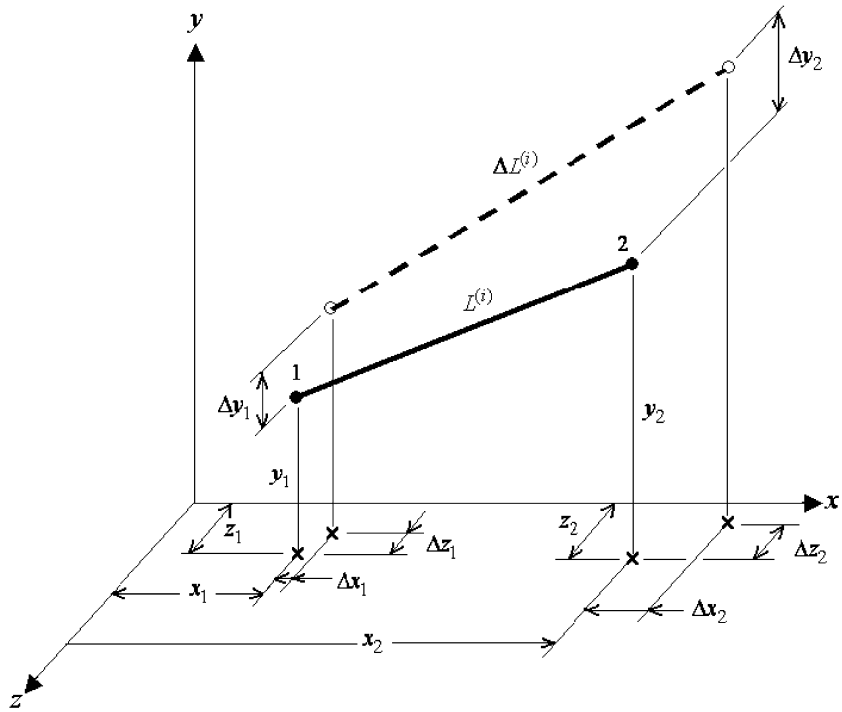

Fig. B1. Nonlinear 3-D truss variables.

$$uy_{21} = uy_2 - uy_1 \tag{B.20}$$

$$uz_{21} = uz_2 - uz_1 \tag{B.21}$$

The term xli is the change in the original length, xl, of the ith member, in the element coordinate system. The terms ux_{21}, uy_{21} and uz_{21} are the change in displacements at the ends of the element.

Now, using the Green strain equation

$$\varepsilon = \frac{xli^2 - xl^2}{2xl^2} \tag{B.22}$$

and substituting in the original and new truss lengths in matrix form yields

$$\varepsilon = \frac{(\vec{xl}_{21} + \vec{p}_{21})^{\mathrm{T}}(\vec{xl}_{21} + \vec{p}_{21}) - (\vec{xl}^{\mathrm{T}}_{21}\vec{x}_{21})}{2\vec{xl}^{\mathrm{T}}_{21}\vec{xl}_{21}} \tag{B.23}$$

This can be simplified in matrix notation and written as

$$\varepsilon = \vec{xl}^{\mathrm{T}}_{21}\vec{p}_{21} + \tfrac{1}{2}\vec{p}^{\mathrm{T}}_{21}\vec{p}_{21} \tag{B.24}$$

The first portion of Eq. (B.24) represents the linear strain, and the last part of the equation represents the nonlinear contribution. Thus the internal force in the truss is

$$AN = EA\varepsilon \tag{B.25}$$

B.3. Nonlinear 3-D truss element stiffness

The geometric stiffness can now be defined as

$$K_{\mathrm{G}} = \begin{bmatrix} \frac{AN}{xl} & 0 & 0 & -\frac{AN}{xl} & 0 & 0 \\ & \frac{AN}{xl} & 0 & 0 & -\frac{AN}{xl} & 0 \\ & & \frac{AN}{xl} & 0 & 0 & -\frac{AN}{xl} \\ & & & \frac{AN}{xl} & 0 & 0 \\ & & & & \frac{AN}{xl} & -\frac{AN}{xl} \\ & & & & & \frac{AN}{xl} \end{bmatrix} \tag{B.26}$$

where

$$\frac{AN}{xl} = \frac{EA}{xl}\left[\frac{(ux_2 - ux_1)}{xl} + \frac{(uy_2 - uy_1)}{xl} + \frac{(uz_2 - uz_1)}{xl} + \frac{(ux_2 - ux_1)^2}{2xl^2} + \frac{(uy_2 - uy_1)^2}{2xl^2} + \frac{(uz_2 - uz_1)^2}{2xl^2}\right] \tag{B.27}$$

The total nonlinear 3-D truss-element stiffness can now be written by adding the corresponding members of Eq. (B.27) with those in Eq. (B.6) which can be reduced until the final tangent stiffness becomes

$$K_{\mathrm{T}} = \frac{EA}{xl}\left\{1 + \frac{[3(ux_2 - ux_1)]}{xl} + \frac{[3(uy_2 - uy_1)]}{xl} + \frac{[3(uz_2 - uz_1)]}{xl} + \frac{[3(ux_2 - ux_1)^2]}{2xl^2} + \frac{[3(uy_2 - uy_1)^2]}{2xl^2} + \frac{[3(uz_2 - uz_1)^2]}{2xl^2}\right\} \tag{B.28}$$

This is the tangent stiffness that is used for a nonlinear truss element in the developed finite-element algorithm. The performance of the truss element was tested by Hrinda [13] against the snap-through

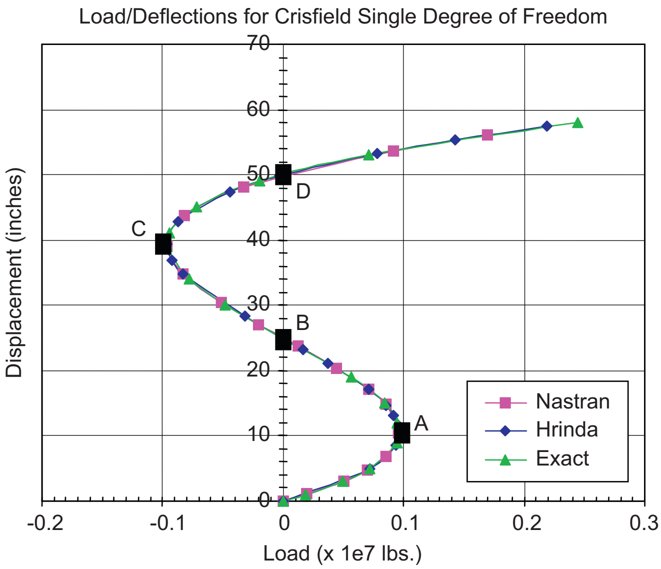

Fig. B2. Load increment vs. displacement for the sdof problem given in Crisfield [6]. The plot shows three equilibrium paths that are closely the same.

Table B1
Quantitative results comparison of sdof problem using nastran, the finite-element program by Hrinda [13] and the exact solution given by Crisfield [6]

Method	A	B	C	D
Nastran	0.0935, 12.59	0.000, 25.00	−0.0962, 39.26	0.0077, 50.38
Hrinda	0.0948, 12.00	0.000, 24.97	−0.0961, 39.00	0.0070, 50.34
Exact	0.0948, 12.00	0.000, 24.82	−0.0962, 39.26	0.0077, 50.38

and snap-back problems that were identified by Crisfield [6,7]. The criteria for a desirable element were the capability to accurately trace an equilibrium path and the capability to reach and pass the critical-limit points within an arc length solution. In Ref. [13], finite element analysis of the Crisfield snap-through and snap-back models were preformed using the tangent stiffness given by (B.28) and Nastran. The load–displacement plots were generated and compared with Crisfield. The results in Ref. [13] showed that the truss tangent stiffness given by (B.28) was behaving properly for large geometric nonlinear responses. An example of the results comparison from Ref. [13] is shown in Fig. B2.

A numerical comparison of the four points shown in Fig. B2 are listed below in Table B1.

Appendix C

C.1. SED formulation

The strain in a truss element was expressed in Eq. (B.22) as

$$\varepsilon = \frac{xl_i^2 - xl^2}{2xl^2} \tag{C.1}$$

where xl is the original length along the local x-axis and xl_i is the new length of the ith member. The formulation is referred to as "Green's strain".

The length change of the element may be large; however, the strains are considered small enough to allow a linear stress strain relationship. The work that is done by external forces to cause deformation is stored within the truss in the form of strain energy. In a load–displacement profile, the area under the curve is the strain energy SE and can be written as

$$SE = \tfrac{1}{2}P\Delta \tag{C.2}$$

where P is the internal load and Δ is the axial displacement. Substituting $P = AE(xli - xl)/xl$ for a truss element yields

$$SE = \frac{1}{2}\frac{AE}{xl}(xl_i - xl)^2 \tag{C.3}$$

Here, A is the truss cross-sectional area and E is Young's modulus. The relationship can also be written in the form

$$SE = \frac{AE(xl)}{2}\left(\frac{xl_i - xl}{xl}\right)^2 \tag{C.4}$$

An alternative form of writing Eq. (C.1) is

$$\left(\frac{xl_i - xl}{xl}\right)^2 = \left(\varepsilon - \frac{xl_i - xl}{xl}\right)2 \tag{C.5}$$

This expression is substituted into Eq. (C.4) to give the strain energy for an ith element which is

$$SE_i = AE(xl)\left(\varepsilon_i - \frac{xl_i - xl}{xl}\right) \tag{C.6}$$

SED is the amount of strain energy per unit volume. The SED can be found by dividing the strain energy by the volume of an element and can be written as

$$SED = \frac{SE}{(xl)A} \tag{C.7}$$

This relationship is modified for use in an optimum criterion by dividing the strain energy by the mass density ρ_i for an ith element to yield

$$SED_i = \frac{SE_i}{\rho_i xl_i A_i} \tag{C.8}$$

To further develop an optimality criterion for stability, a discussion of potential energy follows.

C.2. Optimality criterion

The minimum weight optimization of a truss system is defined as

$$W = \sum_{i=1}^{n} \rho_i A_i xl_i \tag{C.9}$$

where ρ_i is the material density of the ith element, A_i is the cross-sectional area of the ith element and xl_i is the new deformed length of the ith element with n number of elements.

The formal optimization problem can be defined as

$$\text{Find} \quad \{A_i\} \tag{C.10}$$

$$\text{to minimize} \quad \sum_{i=1}^{n} \rho_i A_i xl_i \tag{C.11}$$

$$\text{subject to} \quad PE - \overline{PE} = 0 \tag{C.12}$$

This minimum weight equation is subject to the equality constraint

$$g = PE - \overline{PE} = 0 \tag{C.13}$$

where PE is the total potential energy of a truss system and $\overline{PE}$ is the target total potential energy of the optimal design at the critical nonlinear limit load. This equality constraint has the smallest cross-sectional area that is capable of sustaining an applied load just before the structure becomes unstable. This places a constraint on the limit load that is the same as constraining the total potential energy of the structure. For multiple members, the potential energy may be written as

$$PE = \sum_{i=1}^{\text{ele}} SE_i - \sum_{j=1}^{\text{nodes}} W_j \tag{C.14}$$

where SE_i is the strain energy in the ith element and W_j is the work that is produced from an external force. The relationship can also be written for a structural system that consists of n number of truss elements as

$$PE = \sum_{i=1}^{n} SE_i - \vec{u}^{\mathrm{T}} \overrightarrow{F} \tag{C.15}$$

where SE_i identifies the strain energy in the ith element, $\vec{u}^{\mathrm{T}}$ is a vector transpose of the global displacements and $\overrightarrow{F}$ is the applied design load vector.

The constrained minimization of Eq. (C.9) with the equality constraint is summarized as

$$\text{Find} \quad \{A_i\} \tag{C.16}$$

$$\text{to minimize} \quad \sum_{i=1}^{n} \rho_i A_i x l_i \tag{C.17}$$

$$\text{subject to} \quad PE - \overline{PE} = 0 \tag{C.18}$$

Lagrange developed the tool for solving optimality problems by developing the Lagrangian function which is written as

$$L = f(x) + \sum_{j=1}^{m} \gamma_j h_j(x) \tag{C.19}$$

$$\text{for finding} \quad x \in R^n \tag{C.20}$$

$$\text{to minimize} \quad f(x) \tag{C.21}$$

$$\text{subject to} \quad h_j(x) = 0, \ \ j = 1, 2, \ldots, m \tag{C.22}$$

where γ_j is called the Lagrange multiplier for the solution (x) that lies in some domain.

Lagrange multipliers find the maximum or minimum of a function of several variables subject to one or more constraints. This method reduces a problem with n variables with k constraints to a solvable problem with $n+k$ variables and no constraints. The method introduces a new, unknown scalar variable, the Lagrange multiplier, for each constraint and forms a linear combination that involves the multipliers as coefficients. The Lagrange multiplier can be thought of as how strong the constraint is working to get a point to a maximum or minimum. The objective is to find the conditions for some implicit function so that the derivative, in terms of the independent variables of a function, equals zero for some set of inputs. The point x is stationary in space if

$$\frac{\delta L}{\delta x_i} = 0 = \frac{\delta f}{\delta x_i} + \sum_{j=1}^{m} \gamma_j \frac{\delta h_j(x)}{\delta x_i}, \quad i = 1, 2, \ldots, n \tag{C.23}$$

and

$$h_j(x) = 0, \quad j = 1, 2, \ldots, m \tag{C.24}$$

A stationary point merely means that the solution (x) causes the derivative of a function to equal zero which is equivalent to the point at which the function stops increasing or decreasing. In the case presented in this paper, the solution (x) is the value of the load that causes the tangent at that point to be parallel to the x-axis. This is also the point of minimum SED for all members along their equilibrium path. If this formulation is used and applied to minimizing weight in trusses, then the Lagrangian becomes

$$L = \sum_{i=1}^{n} \rho_i A_i x l_i - \gamma (PE - \overline{PE}) \tag{C.25}$$

where γ is the Lagrange multiplier and $(PE - \overline{PE})$ is the difference between the total potential energy of the system PE and the target potential energy $\overline{PE}$ of the optimal design at the critical-limit load. We desire this difference in potential energy to be a minimum. Differentiating with respect to A_i gives

$$\rho_i x l_i - \gamma_1 \left(\frac{\partial PE}{\partial A_i} + \sum_{j=1}^{\text{dof}} \frac{\partial PE}{\partial u_j} \frac{\partial u_j}{\partial A_i} \right), \quad i = 1, 2, \ldots, n \tag{C.26}$$

The potential energy is a function of the set of displacements. If these displacements place the structure in stable equilibrium, then the total potential energy has a "stationary" value and the following can be written as

$$\frac{\partial PE}{\partial u} = \frac{\partial SE}{\partial u} - F = 0 \tag{C.27}$$

This is the partial of the potential energy from Eq. (C.15). Using the zero statement of Eq. (C.27) with Eq. (C.26), the following simplified form is achieved:

$$\rho_i x l_i - \gamma_1 \left(\frac{\partial PE}{\partial A_i} \right) = 0 \tag{C.28}$$

Consider now the potential energy in Eq. (C.15) as a function of the area of the section. A change in the potential energy with respect to a change in the area can be written as

$$\frac{\partial PE}{\partial A_i} = \frac{\partial SE_i}{\partial A_i} - \frac{\partial u_i F}{\partial A_i} \tag{C.29}$$

which is equal to

$$\frac{\partial PE}{\partial A_i} = \frac{SE_i}{A_i} \tag{C.30}$$

This is possible because the nonlinear strain energy is only a function of the design variable.

Placing Eq. (C.30) into Eq. (C.28) yields

$$\rho_i x l_i - \gamma_1 \left(\frac{SE_i}{A_i} \right) = 0 \tag{C.31}$$

This equation can be rewritten as

$$\frac{\gamma_1}{\rho_i x l_i} \left(\frac{SE_i}{A_i} \right) = 1 \tag{C.32}$$

Using the definition of SED, this equation can be given as the SED optimality criterion of

$$\gamma_1 (SED_i) = 1 \tag{C.33}$$

A recurrence formula for updating the area design variable is found by multiplying this formula by $(A_i)^{\beta}$ to give

$$(A_i)^{\beta} = (A_i)^{\beta} (\gamma_1) SED_i \tag{C.34}$$

where β is a parameter that controls the step size and the convergence rate.

Now, taking the βth root on each side

$$(A_i)_{\text{iter}+1} = (A_i)_{\text{iter}} (\gamma_1) (SED_i)^{1/\beta} \tag{C.35}$$

gives the updated area for the next iteration. The use of this equation within the arc length method will set $1/\beta$ equal to the load increment multiplier λ_i, as shown in Fig. 1. The Lagrange multiplier γ in Eq. (C.35) is set equal to

$$\gamma_i = \frac{\sum_{i=1}^{n} SED}{\sum_{i=1}^{n} SED^2} \tag{C.36}$$

and is placed into Eq. (C.35) along with λ_i to obtain the recurrence relation

$$(A_i)_{\text{iter}+1} = (A_i)_{\text{iter}} \left(\frac{\sum_{i=1}^{n} SED}{\sum_{i=1}^{n} SED^2} \right) (SED_i)^{\lambda} \tag{C.37}$$

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